RAINFALL PREDICTION PROJECT

Complete explanation of the project report and Python code
ELI5 • Doodle-style flowcharts • Concepts • Results • Viva revision

What this PDF covers

1. Project idea and objectives
2. Dataset and data preparation
3. Visualisation and feature engineering
4. Machine Learning algorithms and training
5. Model evaluation and saved model
6. Important code issues and limitations
7. Viva questions and quick revision

Important scope note: The supplied code demonstrates rainfall data processing and rainfall prediction. Although the wider project title refers to soil erosion prediction, the code shown does not implement a complete soil-erosion prediction model.

1. Project explained in very simple terms

Imagine a farmer deciding when to sow seeds, irrigate a field or prepare for heavy rain. Historical rainfall records can help estimate how much rain may fall. This project uses Python and Machine Learning to study past rainfall patterns and predict rainfall for a region.

In one line: Past rainfall data → clean and analyse it → train a Machine Learning model → estimate rainfall.

Project title in the report

Integrated Environmental Forecasting and Soil Erosion Prediction using Meteorological and Remote Sensing Data.

The broader project concept includes environmental and satellite indicators such as NDVI, soil moisture, land-surface temperature, slope and land use/land cover. However, the implementation in the supplied notebook focuses on rainfall forecasting. Do not claim that the complete erosion model is already implemented.

Doodle-style project workflow

Stage	What happens	Simple meaning
1. Data collection	Load rainfall CSV files	Gather historical records
2. Cleaning	Remove unwanted columns, parse dates, handle missing values	Fix messy data
3. Analysis	Create yearly, monthly and region-wise charts	Understand patterns
4. Feature engineering	Create date, season and previous-rainfall features	Give the model useful clues
5. Model training	Train and compare regression algorithms	Teach the computer
6. Evaluation	Measure prediction errors and R^2	Test the model
7. Saving	Save model as rainfall_model.pkl	Keep the trained model for later use

2. Dataset and data collection

The notebook reads CSV files from a local data folder. The files contain daily rainfall information for eight regions: Bareli, Begumganj, Berasia, Bhopal, Ghairatganj, Raisen, Silvani and Udaipura. Raisen has two source files in the folder, which is why nine CSV files are loaded for eight unique regions.

Notebook output	Value
CSV files loaded	9
Unique regions	8
Initial combined dataframe	1,359 rows × 35 columns
Long-format dataframe	41,356 rows × 3 columns
Columns after reshaping	Date, Region, Rainfall
Missing rainfall values found	1,040

Key code ideas

os.listdir(folder_path) lists files in a folder.
pd.read_csv() reads each CSV file into a dataframe (a table).
pd.concat() combines the tables into one dataframe.
df.head() displays the first few rows.

The code skips the first two metadata rows and uses on_bad_lines='skip' to skip malformed lines while reading. It extracts the region name from each filename and adds it as a Region column.

3. Data cleaning and reshaping

The original rainfall table stores each day in a separate column: DRF01, DRF02, DRF03 … DRF31. The notebook uses pandas melt() to convert those day columns into individual rows. It then extracts the day number and combines YEAR, MONTH and Day into a Date column.

Before reshaping (simplified)	After reshaping (simplified)
YEAR=1991, MONTH=6, DRF01=0, DRF02=0	1991-06-01 | Bareli | 0.0
YEAR=1991, MONTH=6, DRF03=2.8	1991-06-03 | Bareli | 2.8

Important functions: melt() changes wide data to long data; pd.to_datetime() creates dates; dropna(subset=['Date']) removes invalid dates; fillna(0) replaces missing rainfall entries with zero.

Interpretation warning: A missing rainfall value is not necessarily the same as zero rainfall. Replacing all missing values with zero can affect the model and should be justified against the source-data meaning.

4. Exploratory data analysis and graphs

The notebook creates four main visualisations:

Graph	What it tells you
Horizontal bar chart	Compares total rainfall across regions.
Yearly line chart	Shows how total rainfall changes from year to year.
Monthly line chart	Shows average rainfall by month.
Month-versus-region heatmap	Shows how average rainfall differs across months and regions.

These charts help identify seasonal patterns, differences between regions and changes over time. The PDF code listing shows the plotting commands, but the actual graph values should be read from the generated notebook plots.

5. Preparing features for Machine Learning

A feature is an input clue given to a Machine Learning model. The target (y) is the value the model is trying to predict. In this project, the target is Rainfall.

Feature	Meaning
Year	Calendar year of the observation.
Month	Month number (1–12).
Day	Day of the month.
Region / Region_encoded	Which region the record belongs to, represented numerically or with indicator columns.
Season	The code labels June–September as monsoon (1) and other months as non-monsoon (0).
Lag1, Lag2, Lag3	Rainfall from one, two and three previous observations for that region.
Rolling3	Three-observation rolling average calculated within each region.

ELI5 example: If the previous three rainfall values are 10, 20 and 30 mm, their average is (10 + 20 + 30) ÷ 3 = 20 mm. A rolling feature gives the model a summary of recent rainfall.

The code sorts by Region and Date before creating lag and rolling features. It drops rows that have missing lag/rolling values because the earliest observations do not have enough previous records.

6. Machine Learning: training and testing

The notebook first tests a basic set of calendar and region features, then later builds a richer time-series feature set using season, lag and rolling features.

Training and testing split

The final model uses the first 80% of the ordered dataset for training and the remaining 20% for testing. Training data teaches the model; testing data checks predictions on later records that were held back.

Algorithms compared

Algorithm	ELI5 explanation
Linear Regression	Fits a simple mathematical relationship between inputs and the target.
Decision Tree Regressor	Uses a sequence of decision rules to produce a numerical prediction.
Random Forest Regressor	Combines predictions from many decision trees.
K-Nearest Neighbours (KNN)	Uses nearby/similar examples to estimate a value.
Support Vector Regression (SVR)	Fits a function within a chosen error tolerance.
Gradient Boosting Regressor	Builds a sequence of models, with later models improving on earlier errors.

Random Forest is the top model in the notebook's initial comparison by R^2 score. The final version uses 300 trees and a maximum depth of 12.

Final model configuration

```
rf_model = RandomForestRegressor(n_estimators=300, max_depth=12, random_state=42)
```

n_estimators=300: the forest contains 300 decision trees.
max_depth=12: limits the depth of each tree.
random_state=42: makes the randomised process repeatable.

7. Understanding evaluation metrics

Metric	Simple meaning	How to interpret
MAE	Average absolute difference between actual and predicted rainfall.	Lower is better. Same unit as rainfall.
MSE	Average of squared prediction errors.	Lower is better; large errors are penalised more.
RMSE	Square root of MSE.	Lower is better; same unit as rainfall.
R^2	How much variation in the target is explained by the model.	Closer to 1 is generally better; can be negative.

Reported final-model results in the code

Metric	Reported value
MAE	0.4342408408
MSE	5.7019626194
RMSE	2.3878782673
R^2	0.9740771906 (about 97.41%)

These are the notebook's printed results. They should be treated as preliminary until the data leakage concern described in the next section is corrected and the model is evaluated again.

8. Feature importance

The final Random Forest reports the following leading feature-importance values:

Feature	Importance	Approximate share
Rolling3	0.631294	63.13%
Lag1	0.203534	20.35%
Lag2	0.151746	15.17%
Day	0.003784	0.38%
Year	0.002812	0.28%
Lag3	0.002448	0.24%
Month	0.001321	0.13%

The model relies most heavily on the rolling average and recent rainfall observations. Feature importance describes how the fitted forest used features; it does not prove that those features cause rainfall.

9. Saving the trained model

The notebook uses Joblib to save the trained Random Forest model in a file named rainfall_model.pkl.

```
joblib.dump(rf_model, "rainfall_model.pkl")
```

This lets a later Python program load the saved model and use it without repeating the training process. Saving a model does not automatically create a website or live forecasting service; those would need additional application code.

10. Important technical issues to know

These points matter when explaining the results honestly, especially during a project viva.

1. Potential data leakage from Rolling3. The code calculates Rolling3 using the current rainfall value as part of the three-observation average. But the current rainfall is the value the model is supposed to predict. This allows information from the answer to enter the input features, which can make test performance look much better than it really is. A safer feature for predicting the current observation is a shifted rolling average based only on earlier observations.

2. Missing values are set to zero. The notebook fills 1,040 missing rainfall values with zero. This assumes missing means no rainfall, which may not be true for every record. Check the source data documentation before making that assumption.

3. The earlier and final model scores differ. The basic model with only Year, Month, Day and encoded Region gives much weaker results. The later model includes lag and rolling features and reports a much higher R^2. The two results come from different feature sets and evaluation setups, so they should not be presented as the same experiment.

4. Sequential split is important. The final code overwrites the earlier random split with a chronological 80:20 split. This is more suitable for time-series evaluation, but the ordering and duplicate dates across regions should still be reviewed carefully.

5. Prediction example needs care. The notebook's earlier simple Linear Regression example predicts 3.4617 for Bhopal on 1 July 2025. That example belongs to the earlier model, not the final Random Forest model with lag features. Do not describe it as the final model's prediction.

11. Project scope, limitations and future work

What the supplied implementation shows

The code demonstrates rainfall data ingestion, reshaping, cleaning, visualisation, feature creation, comparison of regression algorithms, Random Forest training, evaluation, feature importance and model saving.

What is not demonstrated in this code

A complete soil-erosion prediction workflow is not implemented in the supplied notebook. The broader project idea mentions remote sensing and environmental factors, but this code does not show a trained erosion model, erosion-risk map or integration of satellite-derived variables such as NDVI, slope, soil moisture and land-use/land-cover.

Limitations and possible future improvements

Limitation / gap	Possible next step
Only eight regions in the supplied data	Add more locations and test generalisation.
Missing values replaced with zero	Validate missing-data meaning and use an appropriate strategy.
Potential leakage in Rolling3	Shift rolling features so they use only past rainfall.
Historical data only	Evaluate a live weather-data feed as a separate extension.
No full erosion model in this code	Add validated GIS/remote-sensing features and train/evaluate an erosion model.
No user-facing application shown	Build a dashboard or application only after validating the model.

The wider report discusses applications such as agricultural planning, irrigation, water management, flood/drought preparedness, environmental conservation and land management. These are potential uses; practical deployment requires further validation.

12. Viva questions and short answers

Question	Short answer
What is the project about?	Using historical rainfall data and Machine Learning to estimate rainfall.
Which language is used?	Python.
Which library handles tabular data?	Pandas.
What is a dataframe?	A table-like data structure in Pandas.
Why use melt()?	To convert day-wise columns into rows.
What is feature engineering?	Creating useful input variables from raw data.
What is a lag feature?	A previous observation used as an input.
What is Rolling3?	A three-observation rolling average in this notebook.
Which model is selected?	Random Forest Regressor.
Why compare multiple algorithms?	To see which performs better under the chosen evaluation.
What is MAE?	Mean absolute prediction error.
What does R^2 represent?	A measure of how well the model explains variation in the target.
Why use Joblib?	To save and reload the trained model.
What is the saved file?	rainfall_model.pkl.
Does this code predict soil erosion?	No. The supplied notebook implements rainfall prediction, not a complete erosion model.

13. One-minute project explanation

"Our project uses historical daily rainfall records from eight regions. We load and combine the CSV files using Pandas, convert the original day-wise format into a Date–Region–Rainfall table, clean the data and analyse rainfall patterns through graphs. We then create features such as year, month, season, previous rainfall values and a rolling average. We compare different regression algorithms and train a Random Forest model. The notebook reports MAE, MSE, RMSE and R^2, calculates feature importance and saves the trained model as rainfall_model.pkl using Joblib. The current code demonstrates rainfall forecasting; the complete soil-erosion prediction component remains future work."

14. Final quick revision sheet

Remember	Answer
Input data	Historical rainfall CSV files
Locations	8 regions; 9 CSV files loaded
Final table	Date, Region, Rainfall
Total records	41,356 before lag-feature rows are dropped
Main libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib
Selected model	Random Forest Regressor
Final model settings	300 trees, max depth 12, random state 42
Reported metrics	MAE 0.4342; MSE 5.7020; RMSE 2.3879; R^2 0.9741
Most important feature	Rolling3 (0.631294)
Saved file	rainfall_model.pkl

Remember	Answer
Main caution	Rolling3 includes current rainfall in the shown code, creating potential leakage.
Implementation boundary	Rainfall prediction is implemented; full soil-erosion prediction is not shown.

Prepared from the supplied project report explanation and the 15-page Python code PDF. Numerical results and implementation descriptions are reproduced as shown in the supplied code; technical cautions are clearly identified separately.